Gunnar König

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Personal Data

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Academic Positions

2023–present	**Postdoctoral Researcher**, Tübingen AI Center, University of Tübingen Supervisor: Prof. Ulrike von Luxburg, Theory of Machine Learning Group. Topics: Explainable AI, Causal Inference, Foundation Models.
2026	**Research Stay**, University of Vienna Host: Prof. Moritz Grosse-Wentrup. Topic: Explainable AI.
2025	**Research Stay**, University of Amsterdam Host: Prof. Tim van Erven, Korteweg-de Vries Institute for Mathematics. Topic: Algorithmic recourse.
2024	**Research Stay**, Leibniz Institute for Prevention Research and Epidemiology – BIPS, Bremen Host: Prof. Marvin N. Wright. Topic: Feature importance for scientific inference.
2019–23	**Affiliated Researcher**, University of Vienna Group of Prof. Moritz Grosse-Wentrup.

Education

2019–23	**PhD in Machine Learning**, MCML, LMU Munich Supervisors: Prof. Bischl (LMU), Prof. Grosse-Wentrup (University of Vienna) Thesis: "If interpretability is the answer, what is the question?" Committee: Prof. Bischl (LMU), Prof. Augustin (LMU), Prof. Lipton (CMU). Topics: Explainable AI, Causal Inference.
2016–18	**M.Sc. in Data Science**, Elitenetzwerk Bayern, LMU Munich Thesis: "Causal Relationships in Calcium Imaging" Topics: Causal Inference, Deep Learning, Medical Data.
2015	**Exchange semester**, École Polytechnique de Montréal Topics: Deep Learning, Biomedical Data Analysis.
2012–16	**B.Sc. in Information Systems**, TU Munich Thesis: "Online Classification Using Adaptive Sparse Grid Kernels" Topics: Machine Learning, Algorithms, Entrepreneurship. Additional coursework from **B.Sc. Maths** (2015–16).
2012	**Abitur**, Friedrich-List-Gymnasium Reutlingen Awards for outstanding grades in maths, physics, chemistry, and for the overall grade (1.0). Special award for social outreach against discrimination.

Selected Talks

2025–26	**Disentangling Interactions and Dependencies in Feature Attribution** Computer Science Colloquium, University of Vienna (2026); Theory of Interpretable AI Seminar, Host: Tim van Erven (University of Amsterdam, 2025); Interpretable Machine Learning Seminar, Online, Hosts: Giuseppe Casalicchio (LMU Munich) and Marvin Wright (BIPS Bremen, 2025); Stochastics Seminar, University of Amsterdam, Host: Tim van Erven (2025)
2023	**Improvement-Focused Causal Recourse** Research Seminar Machine Learning Theory, Host: Ulrike von Luxburg (University of Tübingen)
2023	**Improvement-Focused Causal Recourse** AAAI Conference, Oral Presentation
2022	**Relative Feature Importance** Mind Team Meeting, Host: Bertrand Thirion (Inria Saclay Paris)
2021	**Interpretable Machine Learning – Jack of All Trades, Master of None?** Convey Seminar, Host: Debarghya Ghoshdastidar (TU Munich)

Teaching

Courses that I designed and taught as the main instructor are marked with †.

2026	**Statistical Machine Learning**, University of Tübingen Lecture and Exercise, 9 ECTS, Master; coordinator of the tutorials. Together with Ulrike von Luxburg.
2026	**Explainability in STEM Fields**†, AI Winter School, Austrian Academy of Sciences, Vienna 4h tutorial, researchers from all disciplines.
2024	**Explainable Machine Learning**†, University of Tübingen Seminar, 3 ECTS, 20 students, Master level. Together with Ulrike von Luxburg, Sebastian Bordt, Robi Bhattacharjee.
2022	**Multivariate Statistics**, LMU Munich Exercise, 2 ECTS, 20 students, Master; tutor and exam. Together with Giuseppe Casalicchio.
2022	**Advanced Machine Learning**, LMU Munich Lecture and Exercise, 6+3 ECTS, Master; lecture material and exercises. Together with Giuseppe Casalicchio, Viktor Bengs, Susanne Dandl.
2022	**Introduction to Causal Inference**†, LMU Munich Seminar, 6 ECTS, Bachelor. Together with Fabian Scheipl, Susanne Dandl, Moritz Herrmann.
2021	**Explainability and Causality**†, Austrian Academy of Sciences, Vienna 2-day tutorial, PhD students. Together with Moritz Grosse-Wentrup.
2021	**Tutorial Explainability and Causality**, ZENITH doctoral programme, Online 2-day tutorial, PhD students; exercise. Together with Moritz Grosse-Wentrup.

2021 **Philosophy and Explainable AI**[†], LMU Munich
Seminar, 6–9 ECTS, 30 students, Master.
Together with Timo Freiesleben.

2021 **Seminar Causality and Graphical Models**[†], LMU Munich
Seminar, 6–9 ECTS, 15 students, Master.
Together with Susanne Dandl, Timo Freiesleben.

2021 **Current Research in Data Science**, LMU Munich
Seminar, 9 ECTS, 20 students, Master.
Together with Giuseppe Casalicchio.

2020 **Seminar Ethics in AI**[†], LMU Munich
Seminar, 6–9 ECTS, 15 students, Bachelor and Master.
Together with Susanne Dandl, Christoph Molnar, Florian Pfisterer.

2020 **Seminar Current Research and Applications in Interpretable Machine Learning**, LMU Munich
Seminar, 9 ECTS, 20 students, Master.
Together with Giuseppe Casalicchio, Christoph Molnar, Julia Herbinger.

2020 **Seminar Causality and Graphical Models**[†], LMU Munich
Seminar, 6–9 ECTS, 10 students, Master.
Together with Christoph Molnar, Susanne Dandl.

2019 **Seminar Limitations of Interpretable Machine Learning**, LMU Munich
Seminar, 6–9 ECTS, 10 students, Master.
Together with Christoph Molnar, Giuseppe Casalicchio, Christian Scholbeck, Florian Pfisterer.

Supervision

2026–present **Karolin Frohnapfel**, PhD Student
Topic: Attention as Explanation in Tabular Foundation Models

2026–present **Karl Gauck**, BSc Student
Topic: Explanation Cards for SHAP

2026–present **Benedikt Bleidt**, BSc Student
Topic: Explanation Cards for Counterfactual Explanations

2024–present **Eric Günther**, PhD Student
Topics: Disentangling Interactions and Dependencies in Feature Attribution, published at AISTATS 2025; Explanation Cards, published at AIES 2026; Finding Interpretable Structure in Complex Models (ongoing)

2024 **Anna Vollweiter**, Master Thesis
Topic: Long-term Robustness of Recourse

2024 **Selina Mail**, Bachelor Thesis
Topic: Feature Importance for Conditional Independence Testing

2023–24 **Fiona Ewald**, PhD Student
Topic: A Guide To Feature Importance Methods for Scientific Inference, published at XAI 2024

2023 **Rebekka Schade**, Bachelor Thesis
Topic: Algorithmic Recourse in Practice

2021–23 **Christoph Luther**, Master Thesis, then PhD Student
Topic: Efficient SAGE Estimation via Causal Structure Learning, published at AISTATS 2023

2020 **Valentyn Melnychuk**, Master Thesis
Topic: Conditional Normalizing Flows for Feature Importance

Community Service and Social Outreach

2026 **Co-Organizer**, Reflections on XAI: Are We Focusing on the Right Fundamental Challenges?
Workshop at the ELLIS UnConference 2026, Paris.
Together with Nicolò Cesa-Bianchi, Emmanuel Esposito, Ulrike von Luxburg, Maximilian Thiessen.

2025 **Co-Organizer**, Workshop on the Theory of Explainable Machine Learning
Co-located with the ELLIS UnConference and EurIPS, Copenhagen.
Together with Robi Bhattacharjee, Nicolò Cesa-Bianchi, Tim van Erven, Emmanuel Esposito, Ulrike von Luxburg.

2019–present **Reviewing** for AISTATS, FAccT, NeurIPS, ICML, ICLR, TMLR, Nature.
Top Reviewer Award for NeurIPS 2025

2018–22 **Co-Founder and Vice President**, ThinkTech (prev. ConsciousCoders)
I helped start and grow the nonprofit to more than 30 members with computer science, law, politics, design, and philosophy backgrounds. Among others, we taught university lectures and created workshops for schools, industry, and policymakers. My highlights were the side events we hosted at the United Nations in Geneva to inform about the dangers of autonomous weapon systems.

2018–22 **Co-Founder**, WeRePack.org
Plugin that helps online shops reuse packaging and reduce waste; founded as part of the SEA:certificate (Social Entrepreneurship Academy), won the Strascheg Award.

Non-academic Positions

2017–23 **Freelance Data Scientist**
Led and grew an international team of data scientists to build a deep-learning-based disease diagnosis system for a medical startup.
Gave lectures introducing ML fundamentals to industry practitioners.

2016 **Data Scientist**, Siemens, Munich
Using Machine Learning to predict financial indexes for in-house customers. The systems are now used in production.

List of Publications

1,500+ citations (Google Scholar), h-index: 12. * denotes shared first authorship.

[1] E. Günther, B. Szabados, K. Meding, G. König, S. Bordt, and U. von Luxburg. We need expla-

nation cards to connect explanation algorithms to the real world. In *AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, 2026.
[2] G. König, M. Pawelczyk, U. von Luxburg, and S. Bordt. Validity threats for foundation model research. *arXiv preprint arXiv:2606.05029*, 2026.
[3] T. Freiesleben, K. Meding, and G. König. Explainable AI isn't enough! Rethinking algorithmic contestability. In *AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, 2026.
[4] G. König, H. Fokkema, T. Freiesleben, C. Mendler-Dünner, and U. von Luxburg. Performative validity of recourse explanations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
[5] G. König*, E. Günther*, and U. von Luxburg. Disentangling interactions and dependencies in feature attribution. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025.
[6] F. K. Ewald, L. Bothmann, M. N. Wright, B. Bischl, G. Casalicchio, and G. König. A guide to feature importance methods for scientific inference. In *World Conference on Explainable Artificial Intelligence (XAI)*, 2024.
[7] G. König*, S. Dandl*, K. Blesch*, T. Freiesleben*, J. Kapar, B. Bischl, and M. N. Wright. CountARFactuals – generating plausible model-agnostic counterfactual explanations with adversarial random forests. In *World Conference on Explainable Artificial Intelligence (XAI)*, 2024.
[8] T. Freiesleben, G. König, C. Molnar, and A. Tejero-Cantero. Scientific inference with interpretable machine learning: Analyzing models to learn about real-world phenomena. *Minds and Machines*, 2024.
[9] C. Molnar, G. König, B. Bischl, and G. Casalicchio. Model-agnostic feature importance and effects with dependent features: A conditional subgroup approach. *Data Mining and Knowledge Discovery*, 38(5):2903–2941, 2024.
[10] G. König, T. Freiesleben, and M. Grosse-Wentrup. Improvement-focused causal recourse (ICR). In *AAAI Conference on Artificial Intelligence (AAAI), oral presentation*, 2023.
[11] G. König*, C. Luther*, and M. Grosse-Wentrup. Efficient SAGE estimation via causal structure learning. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2023.
[12] T. Freiesleben and G. König. Dear XAI community, we need to talk! Fundamental misconceptions in current XAI research. In *World Conference on Explainable Artificial Intelligence (XAI)*, 2023.
[13] G. König. *If Interpretability is the Answer, What is the Question? A Causal Perspective on Explainable Machine Learning*. PhD thesis, LMU Munich, 2023.
[14] G. König*, C. Molnar*, T. Freiesleben*, J. Herbinger, T. Reisinger, G. Casalicchio, M. N. Wright, and B. Bischl. Relating the partial dependence plot and permutation feature importance to the data-generating process. In *World Conference on Explainable Artificial Intelligence (XAI)*, 2023.
[15] C. Molnar, G. König, J. Herbinger, T. Freiesleben, S. Dandl, C. A. Scholbeck, G. Casalicchio, M. Grosse-Wentrup, and B. Bischl. General pitfalls of model-agnostic interpretation methods for machine learning models. In *xxAI – Beyond Explainable AI, ICML 2020 Workshop, Lecture Notes in Computer Science 13200*. Springer, 2022.
[16] G. König, C. Molnar, B. Bischl, and M. Grosse-Wentrup. Relative feature importance. In *International Conference on Pattern Recognition (ICPR)*, 2021.
[17] G. König, T. Freiesleben, and M. Grosse-Wentrup. A causal perspective on meaningful and robust algorithmic recourse. In *ICML Workshop on Algorithmic Recourse*, 2021.

Last updated: October 28, 2026